

ATPL EASA STUDY GUIDE | 010 - AIR LAW

Chapter 9: Holding Procedures & Standards

Syllabus Source: CAE Oxford (pp. 215-226) | Pass Target: >= 90% AviationExam

1. HOLDING PATTERN GEOMETRY & STANDARDS

- Standard Pattern: Racetrack pattern using RIGHT-HAND turns (non-standard pattern uses left-hand turns).
- Rate of Turn: 3° per second (Rate 1 turn) or 25° bank angle, whichever requires less bank.
- Outbound Timing: 1 MINUTE at or below 14,000 ft (4,250 m). 1.5 MINUTES above 14,000 ft.
- Timing Point: Outbound timing begins over or abeam the holding fix, whichever occurs later (or on wings level if abeam point cannot be identified).

2. MAXIMUM HOLDING SPEEDS (PANS-OPS / EASA)

Altitude Band	Normal Maximum Holding Speed	Turbulence Conditions
Up to 14,000 ft	230 kt (Cat A & B: 170 kt)	280 kt (or 0.8 Mach whichever is less)
Above 14,000 ft up to 20,000 ft	240 kt	280 kt (or 0.8 Mach whichever is less)
Above 20,000 ft up to 34,000 ft	265 kt	280 kt (or 0.8 Mach whichever is less)
Above 34,000 ft	0.83 Mach	0.83 Mach

3. THE THREE ENTRY SECTORS (WITH 5° BUFFER)

Entry is determined by the aircraft heading relative to the 3 entry sectors with a 5° flexibility buffer:

- Sector 1 (Parallel Entry - 110° sector): Fly to fix, turn to parallel outbound track on reciprocal heading, fly for 1 (or 1.5) min, turn left towards holding side to intercept inbound track.
- Sector 2 (Offset / Teardrop Entry - 70° sector): Fly to fix, turn to a heading 30° to the holding side, fly for 1 (or 1.5) min, turn right to intercept inbound track.
- Sector 3 (Direct Entry - 180° sector): Fly to fix, turn directly right to follow the holding pattern.

4. HOLDING IN MOUNTAINOUS TERRAIN & WIND DRIFT

[!] EXAM TRAP (AVIATIONEXAM) - Triple Drift Rule & Mountainous MOC

1. Triple Drift Rule: On the outbound leg, apply THREE TIMES (3x) the inbound drift angle in the opposite direction. 2. Mountainous Terrain MOC: Standard MOC in the primary holding area is 300 m (984 ft / 1,000 ft). In designated mountainous areas, the Minimum Obstacle Clearance is doubled to at least 600 m (2,000 ft).

5. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: What is the maximum holding speed under PANS-OPS up to 14,000 ft in normal conditions?

- [A]: 210 kt
- [B]: 230 kt
- [C]: 240 kt
- [D]: 250 kt

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Under PANS-OPS, normal max holding speed up to 14,000 ft is 230 kt (170 kt for Cat A/B).

Q2: When does outbound timing begin in a holding pattern if the abeam position cannot be determined?

- [A]: Passing 90° to the holding track.
- [B]: Immediately upon wings level on outbound heading.
- [C]: 30 seconds after initiating the turn.
- [D]: When crossing the holding radial on DME.

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Outbound timing begins over or abeam the fix, or upon wings level if the abeam position cannot be identified.

Q3: What angular size corresponds to Sector 2 (Offset / Teardrop entry)?

- [A]: 70°
- [B]: 110°
- [C]: 180°
- [D]: 90°

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [A]. Sector 1 (Parallel) is 110°; Sector 2 (Offset) is 70°; Sector 3 (Direct) is 180°.

Q4: If an aircraft applies 4° left drift correction on the inbound leg, what drift correction should be applied on the outbound leg?

- [A]: 4° right
- [B]: 8° right
- [C]: 12° right
- [D]: 16° right

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [C]. Under the triple drift rule, outbound drift is 3 x inbound drift = 3 x 4° = 12° in the opposite direction (right).

Q5: What is the maximum holding speed above 14,000 ft up to 20,000 ft under PANS-OPS in turbulence?

- [A]: 240 kt
- [B]: 280 kt (or 0.8 Mach whichever is less)
- [C]: 300 kt
- [D]: 265 kt

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. In turbulence, maximum holding speed is 280 kt or 0.8 Mach (whichever is less) up to FL 340.

Q6: What is the width of the buffer area extending beyond the boundary of the primary holding area under PANS-OPS?

- [A]: 2.5 NM
- [B]: 5.0 NM
- [C]: 8.0 NM
- [D]: 10.0 NM

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. The buffer area extends 5.0 NM beyond the primary holding area boundary; obstacle clearance tapers to zero at the outer edge.

Q7: What is the maximum holding speed under PANS-OPS above 34,000 ft in all flight conditions?

- [A]: 265 kt
- [B]: 280 kt
- [C]: 0.83 Mach
- [D]: 0.88 Mach

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [C]. Under PANS-OPS, the maximum holding speed above FL 340 is 0.83 Mach.